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Leibnizian Strategies for the Semantical Foundation of the Universal Language

1. As Cassirer (1923) has clearly stated, the historical importance of Leibniz's linguistic thought is mostly due to his *cognitive conception of language*, that is, to his recognition of the constitutive symbolic nature of the human thought.

It is well known that this Leibnizian conception was mainly formulated and developed in the debate against Descartes and within the frame of the epistemological doctrine of the blind thought. In fact, if the principle of evidence always implies a merely instrumental and communicative conception of language, the refusal of such a principle by Leibniz constitutes the very basis of his deep reflection on the importance of symbolism. Symbolism which he considers to be the proper, necessary and essential means of human reason, that which allows us to operate with ideal significations which can only be established by signs and that only signs — as symbols of the virtual determinations of our ideas — enable us to thought. As he says in the celebrated *Dialogus de connexione inter res et verba* of 1677: "I will never be able to know, to discover, to prove without using words or without the presence in my mind of other signs".¹

We are here face to face with one of the passages where this thesis is the most clearly stressed. As a matter of fact, Leibniz's position is not always as strongly stated as in the passage cited above. Compelled by the epistemological

1) "Imo characteres abessent, nunquam quicquam distincte cogitarem, neque ratiocinaremur" (GP 7.191).

and linguistic limits of seventeenth century (cf. Dascal 1978: 174 ff.), as well as by the aporias of his own theory of symbolism (namely the one related to the simple/composed nature of the primitive ideas — since they are simple, indecomposable, they can only be thought by themselves, that is to say, intuitively; but, how is it possible to establish those simple primitive ideas without using definitions which would transform those ideas in composed ideas? (cf. Pombo 1987: 115-119), he has, as if it were, sometimes avoided a strong defense of this thesis. His position is marked by certain ambiguities concerning the effective status conferred to the sign (cf. Pombo 1987: 107 ff.). However, it seems to us to be irrefutable that Leibniz's acknowledgement of the non-merely judicative or demonstrative power of symbolism but also of its prospective and heuristic potentialities — precisely what constitutes the specificity of his theory of symbolism — is the very development of the cognitive conception of language that Leibniz has opposed to the intuitionism of Descartes.

Now, it is precisely this cognitive conception of language which is at the root of Leibniz's project of an Universal Language and, at the same time, of Leibniz' critique of all the proposals of a Universal Language of his predecessors — the line that goes from Llull to Kircher, and that which goes from the English pasigraphers on to Dalgarno and Wilkins.

According to Leibniz, none of his predecessors have managed to guarantee the heuristic virtualities that symbolism may offer (cf. *GP* 7.184) and that for two main reasons, frequently repeated: first, their symbolic systems were not built upon a true and consequent analysis of thoughts, that is to say, they were built on the basis of a set of categories (*predicamenta* or *summa genera*) which can only denounce the analytical insufficiencies of their authors (cf. *GP* 3.216); second, the system of characters and denominations they proposed were arbitrarily, artificially established (cf. *C* 151). For instance, in what concerns Llull's *Ars*, which is an important source of inspiration for the heuristic nature of the Leibnizian project of a Universal Language and whose epistemological value Leibniz fully recognizes (cf. *C* 177 ff.), Leibniz criticises both the arbitrariness of the sign-system that forms the alphabet of the Llullian *Ars*, and his actual list of categories (*GP* 4.61-64; *C* 177). In both cases, what Leibniz can not accept is the absence of an extrinsic foundation of the symbolic system. That is to say, Leibniz does not fall prey to the illusion — to a certain extent present in Llull's project — of an absolute formalism in which the automatical functioning of a set of fixed, operational rules would give the illusion of permitting the progress of science.

As a way to guarantee the prospective and heuristic role of the Universal Language and, simultaneously, to avoid the complete formalism of a well made but empty language, Leibniz considers to be necessary that the symbolic

system would be semantically opened to the reality that it must permit to say. In other words, heuristics in Leibniz also involve a semantic exigency.

Significantly, these two main critiques point towards the two strategies for the semantical foundation of the Universal Language that, according to us, Leibniz has thought out. There are in fact *not one but two* different strategies for the semantical foundation of the Universal Language. The first aim of this paper is precisely to distinguish, to characterize and to relate those two strategies.

2. The *first strategy* appears since the *De arte combinatoria* (1660), where for the first time Leibniz defines the methodological regime that should govern the construction of the Universal Language.

Such a regime, which would essentially be reaffirmed in later formulations, supposes, as its first stage, the determination of all the primitive ideas into which all our concepts can be resolved (cf. *GP* 4.72-73), that is, the exhaustive inventory of all the simple, irreducible terms whose combinatory would permit the production of the totality of the concepts necessary to the full knowledge of reality.

It is easy to understand Leibniz' objective: he aims to protect the cognitive virtualities of language, that is, to guarantee the inventivity of the symbolic system and to avoid the risks of an absolute formalism. He intends to achieve this goal by grounding the symbolic system on the establishment of primitive ideas which can only be thought *out* of language, in the transparency of the self-evidency of the conscience to itself, that is, on intuitive thought (cf. *GP* 4.423), gnosiological correlatum of simple ideas and so postulated ground of all symbolism. Extrinsic foundation is thus conceived here as requiring the return of the symbolic system to the universe of intuition into which primitive ideas can only be thought and, as such, to the world which is to be spoken.

But this strategy implies an enormous, encyclopaedic exigency. An exigency that would involve, as Descartes has pointed out, the previous constitution of the true philosophy, even if not in its entire accomplishment, as Leibniz has contested (cf. *C* 27-28). An exigency that Leibniz himself had not — and could not — been able to solve. It is indeed well known that Leibniz has done various attempts to draw up that inventory — giving examples of numerous series (cf. *GP* 4.70-71, *C* 358-360, *C* 514-515 or *GP* 2.542), reducing it to the idea of God alone (cf. *C* 513), to the idea of God and Nothingness (cf. *C* 430), or finding some pragmatic solution which would avoid the failure of the project, immediately on the level of its initial stage. See, for instance, the distinction established by Leibniz, precisely in the frame of his reply to the apparently un-

surmountable difficulty pointed out by Descartes, between the “termes absolument primitifs” and the “termes primitifs a nostre egard” from which he has admitted one could start out (cf. C 176, C 220-222).

All these various attempts demonstrate the crucial importance Leibniz attributed to this first moment. But it is clear that — as it was inevitable — Leibniz never compiled the definitive, exhaustive list of primitive ideas. However, if this does not permit us to immediately conclude that it was as a result of such insuperable difficulties, that Leibniz had opted for other models of semantic foundation of the Universal Language and developed other lines of research, perhaps it may allow us to face the hypothesis, and even to suggest that, given the insurmountable difficulties to which such a model led, he had tried a second strategy.

This is not the place, nor is it now our aim, to value the difficulties — let us say, the impossibility — of the first strategy or even to establish any causal relationship between the two of them. Here, as well as in other areas of his thought, Leibniz explores multiple perspectives, pursues different lines of research in parallel. What we want to stress is that there is another way of conceiving the semantical foundation of the Universal Language, a *second alternative-strategy*, pursued in parallel throughout his life and which, furthermore, if well succeeded, could reinforce or even substitute the first way mentioned above.

3. We refer to the ‘representativity of the sign’ that Leibniz wanted to conquer for the characteristic signs of the Universal Language to be constructed. That is his aim that — besides operativity or functional capacity of the signs within the formal system — signs should also be “natural”, representative or directly open to the reality they are supposed to say.

However, with this requirement of the representativity of signs that foundation would now be achieved in a quite different way: no longer on an *intelligible* level, by the opening of the system to the ideal universe given by the intuition of the simple primitive ideas (first strategy), but on an immediate, *sensitive* level, by the isomorphism between the characteristic signs and the realities they signify; not then on a primary, intelligible level, but on a sensible level of motivation of signs, or, as we prefer to say, its representativity.

Given the importance of the Leibnizian requirement of the representativity of the characteristic signs, it is easier to understand the heterogeneity of the Leibnizian theory of representativity of the sign. In fact, if one reads carefully, and almost literally, we could say, his many and heterogeneous sum of indications concerned this issue, one can recognize not only the powerful endeavour

Leibniz has done in view of the establishment of such a requirement, as well as the various models of representativity he has indeed pursued, the different ways of gaining such a representativity he has attempted.

As we tried to show elsewhere (Pombo 1987), it is indeed possible to distinguish *three great models* within the Leibnizian theory of the representativity of the sign, which here we can only outline briefly. The *first* model is the extreme proposal of a 'figurative representativity' as the pictural, spacial, imagetic representation of the most pregnants traits of the signified reality — "des petites figures à la place des mots" (GP 5.379), "ce caractere figuré, qui parleroit veritablement aux yeux" (GP 5.379), as Leibniz says in the *Nouveaux Essais*.

In a *second* and weaker model, which we propose to call 'essentialist', representativity is conceived, not as the figuration of the sensory, particular traits of the signified reality as it immediately appears to our perception, but as the direct designation of its essence — "rei ipsius ideam" (GM 5.216) — that is to say, the representation of the essential basis of those particularities, thought out *either* as their causal, generative principle (we are referring to the Leibnizian project of constituting a Characteristic on the basis of a binary arithmetic, since he finds in the latter an admirable similarity between its mode of generation of numbers and the relation of dependence of creatures with respect to God, their ultimate causal principle (cf. FNL 167, C 430) — a good sample of what we propose to call the 'genetic representativity' — the essence of each creature would correspond to its ordered place in the process of creation, then being possible to deduce from each character the totality of properties of the creature it represented as a manifestation of its dependence on the first cause or absolute essence), *or* as the 'key' (*clavis*) or unifying principle of the multiplicity of the properties which compose the signified idea. We are referring to those texts in which Leibniz exposes the project of what he calls a 'characteristica realis' (cf. GP 7.12-13), in which the character would represent less the genesis than the composition of the idea signified, and in which thus representativity implies a process of analysis of the sign itself, which would identify in it, both the constitutive elements of the represented idea and the law of their combination, as it is in the case of the analysis into prime factors (cf. C 277) — 'analytical representativity'.

Finally, in the *third* and last model, Leibniz conceives the representativity as the analogical and structural reproduction of the very system of relations which constitutes the idea and its articulations. Representativity is now conceived as isomorphism, the "rapport constant et réglé" (GP 2.112) between the idea represented and its schematic (geometrical) figuration thus implying a

search, not for similarities (as in the reproductive model of 'figurative representativity') but of "une certaine loy de rapport" (*GP* 1.383).

Within this model, which we propose to call 'expressive', it is yet possible, we believe, to distinguish three subtly different subtypes. Of course, all representativity is in Leibniz expressive since expression is the supreme instance of any type of relation, whether on the metaphysical or the cognitive level. But, precisely because expression is the most ample relation of the system, must we try to analyse it and try to find out its internal differencies. When carefully comparing different passages and fragments such as the *Essais d'analyse grammaticale* (1683/84), the celebrated *Dialogus* (1677) or the *Characteristica Geometrica* (1679), it is indeed fascinating to see how Leibniz is exploring and dealing with slightly different orientations or tendencies of thinking about the possibility of an expressive representativity. In fact, even if the primacy given to structural analogy is what characterizes the expressive model of representativity in general terms, Leibniz conceives structure, *either* as purely formal (the semantic level tending to be entirely reduced to the syntatic — this is the possibility explored in the important fragment *Essais d'analyse grammaticale*; cf. *C* 285), *or* as source of meaning in itself, since it would be able to imitate the real conceived as a structure, (this is the case expounded in the *Dialogus*; cf. *GP* 7.192; where it is structure itself that tends to become semantic), *or* as being absorbed by the sign and through it becoming expressive, as it is suggested in the fragment *Characteristica Geometrica* (cf. *GM* 5.141) where Leibniz clearly recovers the ideal of the semanticity of the sign itself.

Leibniz is thus forced, as if were, to displace the ideal of representativity 'from the sign' (as in the first subtype — which we propose to call 'diagrammatic representativity' — where the sign is emptied of any autonomous signification and reduced to a simple pole of convergence of a multiplicity of relations without any immediate reference to the real universe), to the syntatic structures themselves through its analogy with the supposed structure of the facts (second tendency — structural tendency, properly speaking), and again back to the sign, given that, in the third tendency, the character absorbs the entire expressive function into itself (the 'expressive representativity' — which we now could name 'monadological representativity' — would again come into being between the singular individual (containing relations as its predicates) and the character, as a sign which, through its own flexion, would 'exhibit' the multiple relations of the thing (cf. Pombo 1987: 176-189).

4. Leibniz never explicitly thematizes the differences between the several models of representativity of the sign which, we believe, he has thought out. However, the very fact that he has attempted so many models, explored so many strategies and hesitated, up to the end of his life, with regard to the sign-system to be adopted, can be read as showing how much he aimed to reconcile the rigour guaranteed by the operativity of the sign with the meaning which only representativity can permit, that is, his refusal of the idea of constructing a language that would be totally formalized, and, as such, meaningless.

The representativity of the sign in Leibniz is not then a minor, secondary requirement or a baroque exigency. Yet it is not only a systematic implication decurrent of the principles of 'sufficient reason' and 'pre-established harmony'. It is an issue whose important place within the Leibnizian theory of symbolism must be recognized so that we could understand what determines and mobilizes the two main and apparently divergent directions of Leibniz's linguistic investigations: the study of natural languages and the project of construction of a Universal Language. Only that can make us recognize one possible unifying pole for the three great and apparently contradictory Leibniz's linguistic projects within those three directions:

- a) The study (and perfection) of natural languages (German especially), in which Leibniz looks for that representativity, discovering it both at the level of the etymologies and on the still present motivation of its vocabulary;
- b) the search for the constitution of a Rational Grammar on the basis of the deep universal structure which is expected to underly the grammatical particularities of the various national languages; and
- c) the construction of a Universal Language endowed with a similar or even higher representative nature, and in which the representativity of its signs would somehow repeat (and even perfect) something that had already been achieved in natural languages.

Leibniz's main purpose would be then to investigate the motivated origin of natural languages, to examine the mechanisms responsible for the naturalness of their vocabulary (German especially) and for the deep structure underlying grammatical particularities of the different languages, and to apply those discoveries in the construction of a new philosophical language which, for that reason, Leibniz requires to be endowed of representative signs. In a word, the project of a *characteristica universalis* appears as being closely articulated with the Leibnizian investigations on the natural languages: Leibniz would have explored various levels of motivation in natural languages — from the phenomena of phonetic, rhythmic and articulatory symbolism to onomatopoeia (cf. GP 5.260-264), passing through the search for a direct relation between

sound and idea as referred to in the *Brevis Designatio* (cf. *D* 4.186-189), the relative motivation resulting from the morphological composition of certain words (see the *lux/fero* cases at the *Dialogus*, cf. *GP* 7.192) or the deep universal structure underlying the grammatical particularities of each natural language — tried to comprehend its origin, to understand its determining law, so that he could recover and (re)construct that motivation on the level of the Universal Language, that is, prolong, develop and even perfect the anonymous labour of the popular onomaturgus (since philosophical language would be fundamentally graphic, its motivation would avoid the oscillations and the progressive extinguishment to which natural languages have been submitted by time and human displacements).

That is why, although in complete agreement with Cassirer when he stresses the invaluable importance of Leibniz's cognitive conception of language, we cannot accept Cassirer's view according to which Leibniz's investigations on natural languages are both the negative side of his earlier interest on the *characteristica universalis* and also something which Leibniz did not seem to have valued that much and did not help to make clear (Cassirer 1923: 76-78). On the contrary, we believe that it is precisely because Leibniz has developed a very consistent cognitive conception of language and thus looked for an extrinsical foundation of the symbolic system as a way to guarantee its heuristic virtualities; and furthermore, because he has recognized the insurmountable difficulties to which the model of the semantical foundation through the primitive ideas led that he has intended to recover for the Universal Language the kind of extrinsical foundation he already found at work at the level of natural languages, that is, the second model we have tried to characterize.

Thus, according to this interpretative line of research, the fact that Leibniz has explored so many levels of motivation in natural languages (as briefly referred to above) and, at the same time, pursued so many strategies to conquer the representativity of the characteristic signs (which we here also have tried to distinguish and characterize), shows clearly all the importance he gave to the possibility of reinforcing or even substitute the extremely difficult — let us say, impossible — model of semantical foundation through the primitive ideas by another model which has already shown its efficiency at the level of natural languages. In other words, as it has been already recognized by some scholars (cf. Heinekamp 1972), Leibniz's various, large and profound investigations on natural languages may indeed be thought out as closely articulated with his project of a Universal Language.

5. However, the interest in discussing this subject is not only *internal* to the exegesis of Leibniz's linguistic thought. It is also, we believe, *external* in the sense that this Leibnizian theory of the representativity of the sign (and its various models) may become, in near future, extremely helpful for us to think the question of the representative versus arbitrary nature of the human languages.

In fact, Leibniz's thesis of the motivation of natural languages and the detailed investigations he has developed in this sense, studied among many by Cassirer himself (cf. Cassirer 1923-29, I: 94-111), may appear to have been completely refuted by the modern linguistics which is based, to a large extent, on the intransigent defence of the arbitrariness of the linguistic sign, as established by Saussure in the celebrated *Cours de Linguistique générale*.

But the question, which is not closed even in Saussure (cf. Starobinski 1971), is as old as human reflection on the essence of language and, thus, not easily resolvable. Against the naivité of some professional linguists, who thought to have definitively closed the philosophical question of the origin and nature of human language by proclaiming the arbitrariness of the linguistic sign as the necessary 'epistemological rupture' which has given rise to the new science of linguistics, and given the impossibility, as revealed by the subsequent debates and the multiple conceptual distinctions introduced over more than forty years of polemics — see, for example, the celebrated Benveniste's displacement of the concept of arbitrariness of the sign from the relation signified/signifier to the relation sign/thing (cf. Benveniste 1966: 2.49 ff.) or the more than a hundred titles pertaining to the post-Saussurean debate over the arbitrariness of the sign reviewed by Rudolf Engler (1962; 1964) — of reaching an agreement over the actual nature of this postulate (at least as difficult to support as the opposite position), the thesis of the non-arbitrariness of the sign has reappeared, gaining a surprising degree of consensus among the most distinguished exponents of linguistic sciences.

That is the case of Jakobson, with no doubt one of this century's most eminent analysts of the linguistic phenomena who, in a text significantly called *À la recherche de l'essence du langage*, denounces the total neglect to which some linguists consign the central problem of the connexion between sound and sense, linking it to the negative effect exerted on them by the supposed agreement on Saussure's dogma of the arbitrariness of the sign (cf. Jakobson 1965). Furthermore, the non-conventionalist thesis has been much reinforced by the new evidences issued from naturalistic investigations as, for instance, Karl Bühler's studies (1979) of the different levels — acoustical, vocalic, articulatory, vocabular and phonematic levels — by which the 'pictorial potencial' of human language is expressed, or the studies within the frame of geneticist and

generativist research of, for instance, Lenneberg, looking for a biological matrix for the development of speech and language (cf. Lenneberg 1967), just to refer to two most significant cases. In this sense, too, it is today commonly accepted that, since any language is neither entirely conventional nor entirely motivated, the existence of the three types of motivation (phonetic, morphologic and semantic) is one of the semantic universals (cf. Ullmann 1961).

Only our, let us say, ideological belief in the arbitrariness of language, our dogmatic acceptance of that postulate may induce us to consider the Leibnizian theory of the representativity of the sign as a secondary requirement within Leibniz's linguistic thought. When one looks to the Leibnizian theory of the representativity of the sign, it is then somehow necessary to forget what we know today, or suppose to know.

And Leibniz can help us to think all that such a postulate leaves aside.

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